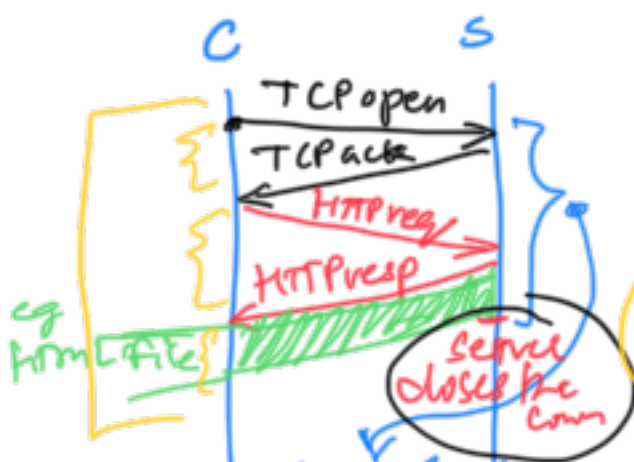


RECAP

A HTTP/1.0 Non-persistent HTTP



Repeat above steps for each object to be requested

NOTE: The objects can be requested in **PARALLEL**

- Too many conn's; Loaded server

$$Time = 2 * RTT + \text{file transfer time}$$

$$+ 2 * RTT + \text{file transfer time of largest file}$$

For 'n' parallel objects

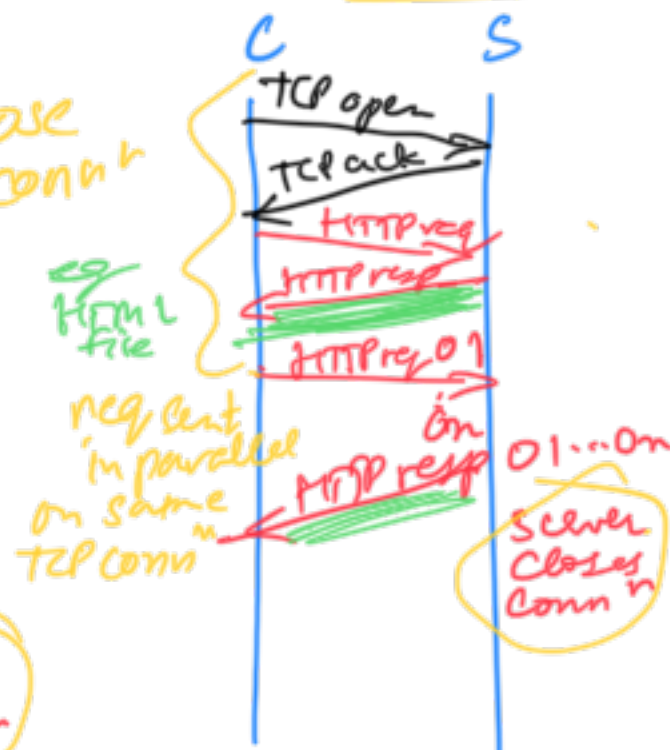
B Persistent HTTP without pipelining



- Single TCP conn for all objects
- requests sent in the order as an object is encountered
↳ cannot send next req until the previous request's response is received
↳ HOL blocking

$$Time = 2 * RTT + t_{file} + n * (RTT + t_{file})$$

C HTTP/1.1 Persistent HTTP with pipelining



- Single TCP conn for all objects
- Req are sent in parallel w/o waiting for prev response

- HOL blocking exists
- solved by HTTP/2.0
- HTTP/3.0 addresses security, congestion control, ... over UDP

extra info

$$Time = 2 * RTT + t_{file}$$

$$+ \left[\frac{n}{\text{pipelining levels}} \right] * (RTT + \text{largest of } (t_{file_1}, t_{file_2}, \dots, t_{file_n}))$$

largest file transfer time

* NOTE: # parallel requests depends on level of parallelism supported by the client browser.

